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Studierichting: Informatica
Oefeningenreeks 5D nr. 1.3

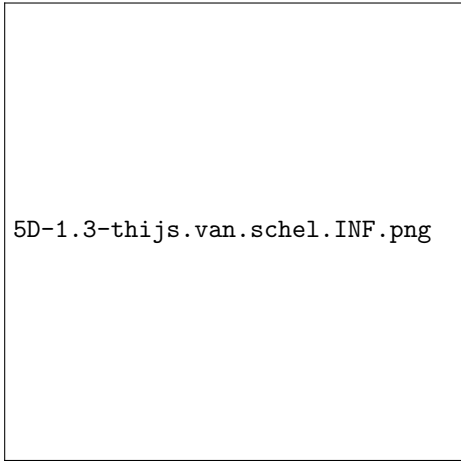
$$\begin{aligned} f(x) = \sqrt[3]{x} &\Rightarrow S = \int_0^8 \left| x^{\frac{1}{3}} \right| \\ &= \left[\frac{3}{4} x^{\frac{4}{3}} \right]_0^8 \\ &= \left(\frac{3}{4} \cdot 8^{\frac{4}{3}} \right) - \left(\frac{3}{4} \cdot 0 \right) \\ &= \frac{3}{4} \cdot 2^4 - 0 = 12 \end{aligned}$$

Tekening:

$$x = 0 \Rightarrow y = 0$$

$$x = 1 \Rightarrow y = 1$$

$$x = 8 \Rightarrow y = 2$$



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References